Calculation (accuracy)

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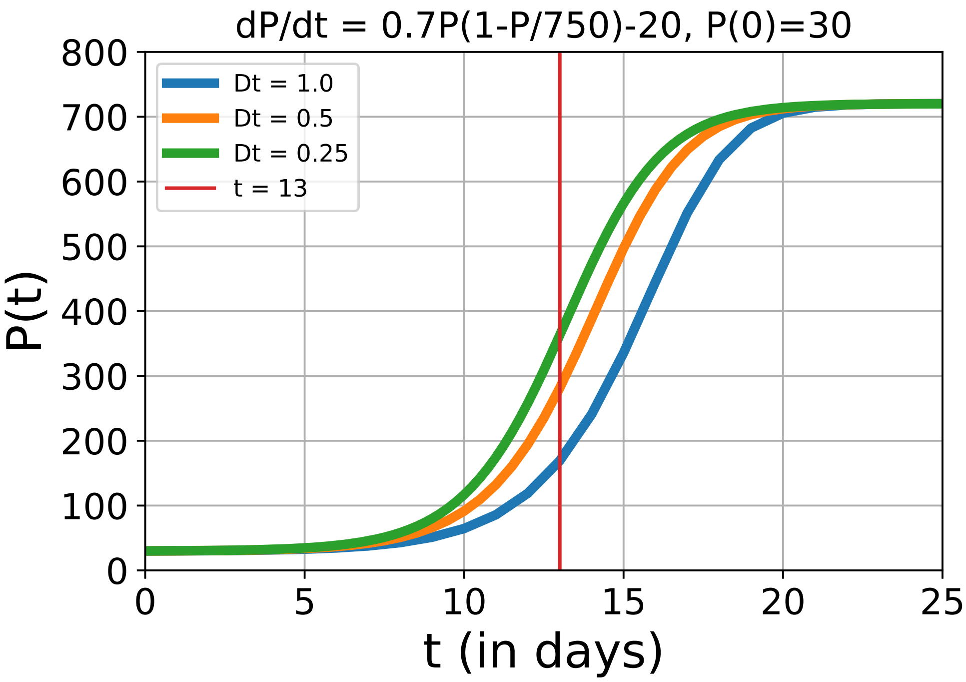
Now you have working code, it is time to investigate the stepsize further.

In the next figure you can see the approximate solutions for $\Delta t = 1$, $\Delta t = \frac{1}{2}$ and $\Delta t = \frac{1}{4}$ day. As you can see the graphs of the biggest two stepsizes differ more than the graphs of the two smallest stepsizes. This behavior is called numerical convergence: The smaller you take Δt, the better your approximation becomes and it will look more and more like the exact solution.



In the figure you can also see that the largest differences in the value of P occur around 13 days:

$\Delta t = 1$ day		$\Delta t = 0.5$ day		$\Delta t = 0.25$ day
$P(13) \approx 169.24$		$P(13) \approx 281.64$		$P(13) \approx 363.09$
	difference: 112.40		difference: 81.45	

So when you want to know the number of fish at 13 days, the stepsize is very important. The difference between $\Delta t = 1$ day and $\Delta t = 0.5$ day is more than 112 fish! And between the two smaller stepsizes, the difference is still more than 81 fish. So the results clearly change a lot, when the stepsize is halved.

Error of an approximation with Euler's method

We will not give any theoretical background here for the estimation of the error of a result with Euler's method. That would go too far for this course. So, without any proof:

> *For differential equation* $\frac{dy}{dt} = f(t, y(t))$ *(and an initial condition), the exact solution at a specific time* t *is* $y(t)$*. With Euler's method and stepsizes* Δt *and* $2\Delta t$*, that exact solution can be approximated by* $w_{\Delta t}$ *and* $w_{2\Delta t}$*, respectively. Then* $E_{\Delta t}$*, the error in* $w_{\Delta t}$*, can be estimated as the difference between* $w_{\Delta t}$ *and* $w_{2\Delta t}$*:*
>
> $$E_{\Delta t} \;=\; y(t) - w_{\Delta t} \;\approx\; w_{\Delta t} - w_{2\Delta t}.$$

So an estimate of the error in the result with $\Delta t = 0.5$ is $281.64 - 169.24 = 112.40$ fish. And an estimate of the error in the result for $\Delta t = 0.25$ is 81.45 fish.

Exercise A

1/1 point (graded)

Now use your program to calculate $P(13)$ with Euler's method for $\Delta t = \frac{1}{8}$.

The *error* of that result can be estimated as:

Use two decimals in your answer.

44.15 ✔ **Answer:** 44.16

44.15

Submit You have used 1 of 3 attempts

Answers are displayed within the problem

Exercise B

1/1 point (graded)

We want the error of $P(13)$ to be smaller than 20 fish (one day's sale). Continue to divide the stepsize by 2 and estimate the error of the result.

The largest Δt for which the estimated error is less than 20 fish is:

Give your answer as a fraction.

1/32 ✔ **Answer:** 0.03125

$\frac{1}{32}$

Answer

Correct:

You are correct, for $\Delta t = 1/32$, $P(13) \approx 440.27$ and the estimated error is $440.27 - 429.34 = 10.93$, which is less than 20.

Submit You have used 3 of 3 attempts

Answers are displayed within the problem

Exercise C

1/1 point (graded)

Now you have found a stepsize that gives the results that are accurate within 20 fish. Use this timestep to solve our problem:

Find t_{eq} such that $P(t_{\text{eq}}) = 720$.

Note that obtaining $P(t_{\text{eq}})$ exactly equal to 720 is not likely. Therefore you should look for the first value in your list of approximations of $P(t)$ that is larger than 720.

Hint: You could use the program line: "indices = np.where(P_arr>=720)".

An approximation of the value of t_{eq} is:

Give your answer with 2 decimals.

24.63

24.63

Submit You have used 3 of 3 attempts

Correct (1/1 point)

Now that you have found a value of t_{eq}, it is time to validate this result and reflect on your findings on the next page.